

7/8

Sixth Semester 2016
Examination: B.S. 4 Years Programme

Roll No. 10024

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Subject: Inorganic Chemistry
Course Code: CHEM-315

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

① Inorganic.

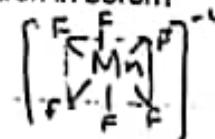
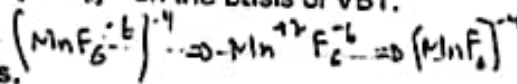
SUBJECTIVE

Q.2 Attempt all short questions

(2 × 10 = 20)

- ✓ What are the limitations of VESPER theory? 143
- ✓ Give comparison of atomic orbital and molecular orbital? 4
- ✓ What is Fajan's rule? 48*
- (✓) Why do some molecules form T shapes instead of trigonal planar shapes? 141*
- ✓ What is hybridization in XeF_6 molecule in solid state? 160
- ✓ What is the significance of the spectrochemical series in coordination chemistry? 282
- (✓) Briefly write the occurrence of actinides in earth crust?
- ✓ What are chelates? Name the factors which influence the stability of chelates.
- ✓ What is the effective nuclear charge experienced by a valence p-electron in boron?
- ✓ What is the structure of $[\text{MnF}_6]^{4-}$ on the basis of VBT.

Note: Attempt all questions.



Q.3. How the molecular orbital theory is applied to explain the structure of transition metal complexes? Draw and discuss the molecular orbital diagram for high and low spin complexes.

Q.4. a) Name the

Monax

b) What is lanthanide contraction and what are its consequences? 458 05

Q.5. Discuss the main postulates of the valence shell electron pair repulsion theory. What are the applications of this theory in determining the shapes of molecules containing four and six electron pairs? 136 10

Limitations of VSEPR Theory

Although the VSEPR theory is very useful in explaining the geometries of a large number of molecules and ions, yet it suffers from the following drawbacks:

- (i) This theory cannot explain the shapes of molecules having any polar bonds e.g. Li_2O should have the same structure and H_2O but in fact it is linear.
- (ii) This theory is unable to explain the shapes of molecules having extensive delocalized π -electron system.
- (iii) This theory is unable to predict the shapes of certain transition metal complexes.
- (iv) This theory also cannot explain the shapes of certain molecules which have inert pair of electrons.



ATOMIC ORBITAL VERSUS MOLECULAR ORBITAL

Q2

Atomic orbital is the region having the highest probability of finding an electron in an atom

Formed by the electron cloud around the atom

The shape is determined by the type of the atomic orbital (s,p,d or f)

Schrodinger equation is used

Monocentric as it is found around a single nucleus

Single nucleus affects the electron cloud

Molecular orbital is the region having the highest probability of finding an electron of a molecule

Formed by the fusion of atomic orbitals that have nearly the same energy

The shape is determined by the shapes of atomic orbitals that make the molecule

Linear combination of atomic orbitals (LCAO) is used

Polycentric as it is found around different nuclei

Two more nuclei affect the electron cloud

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Following are the factors on (which the magnitude of polarizing power of a cation to polarize a nearby anion and the polarisability of an anion depends. These factors have been suggested by Fajans and hence are called Fajan's Rules.

Q 3

1. Charge on Cation or Anion

The cation with higher positive charge attracts the electrons cloud of the anion more strongly towards itself than the cation with smaller positive charge and hence polarizes the anion more strongly. Consequently higher the positive charge on the cation, greater its polarizing power to polarize a given nearby anion. For example in LiCl , BeCl_2 , BCl_3 and CCl_4 molecules, the polarizing power of Li^+ , Be^{2+} and C^{4+} cations (cations of the same period) to polarize Cl^- anion increases with the increase of positive charge on them.

An anion with higher negative charge repels the electron cloud of its outermost shell towards the cation more effectively and hence is polarized by the cation more strongly. it will be polarized by a given cation i.e. more will be its polarizability of C^{4-} , N^{3-} , O^{2-} and F^- anions (anions of the same period) to be polarized by H^+ cation decreases from C_4^{4-} to F^- , since the magnitude of negative charge on these anions also decreases in the same direction.

2. Size of the Cation

The nucleus of a smaller cation is shielded to a lesser extent by the electron shells and, therefore, such a cation attracts the electron cloud of the anion more strongly than a large cation. This means that smaller is the size of the cation, higher is its polarizing power to polarize a given nearby anion. for example in BeCl_2 , MgCl_2 , CaCl_2 , SrCl_2 and BaCl_2 molecules, the polarizing power of Be^{2+} , Mg^{2+} , Sr^{2+} and Ba^{2+} cations (cations of the same group) to polarize Cl^- ion decrease with the increase of their size.

3. Size of the Anion

The electrons of the outermost shell of a large anion are less firmly bound to its nucleus and hence can be withdrawn by the cation more easily. Consequently a larger anion will be polarized by the cation more easily than a smaller anion. This means that larger is the size of the anion more strongly or easily it will be polarized by a given cation i.e., more will be its polarizability. For example in halides of a given cation (e.g. MX halides) the polarisability of halides ions (anions of the same group) will increase from F^- to I^- , since the size of these ions also increases in the same direction.

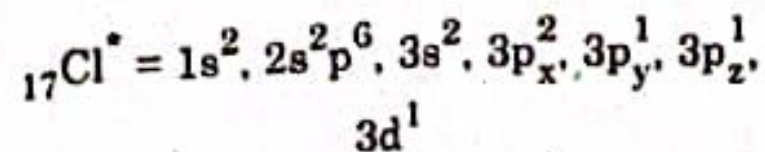
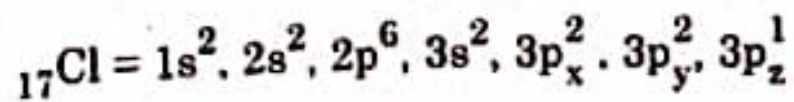
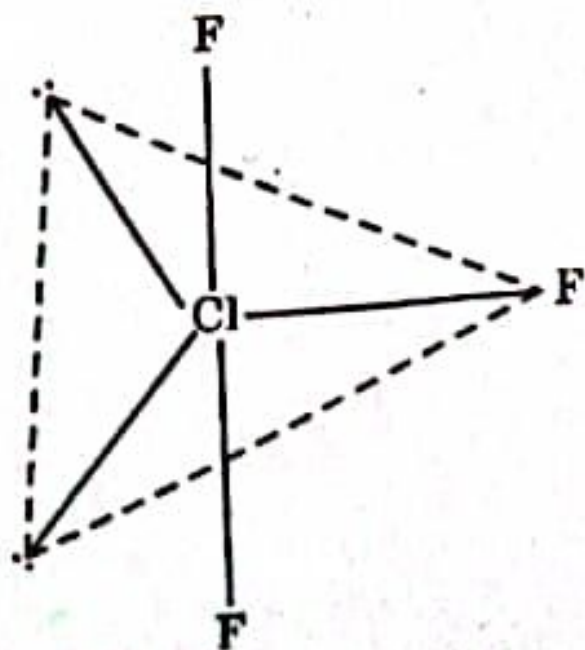
4. Electronic Configuration of the Cation

A cation with 18-electron valence shell configuration ($ns^2p^6d^{10}$ configuration) has a higher polarizing power than that with 8-electron valence shell configuration.

($ns^2p^6d^{10}$ valence-shell configuration because of the fact that the d-electrons of the 18-electron shell shield the nuclear charge of the cation less effectively than the s- and p-electrons of 8-electron shell. Hence the cation with 18-electron valence shell configuration behaves, as if it has greater nuclear charge on it. Greater nuclear charge increase the polarizing power of the cation having 18-electrons valence-shell configuration. Thus Ag^+ cation ($4s^2p^6d^{10}$) has greater polarizing power than K^+ cation ($3s^2p^6$).

9. AB_3E_2 Type

Here A is the central atom, B are the bonded atoms to the central atom and E are the lone pairs of electrons around the central atom. Species of this type have three bonding electron pair and two lone pairs of electrons ($bp = 3$, $lp = 2$). These bonding electron pairs and lone pairs of electrons occupy their positions as far apart as possible so that the electrostatic repulsions between them is minimum. This is most probable if the species of this type has T-shaped bent geometry in which both the lone pairs occupy the equatorial positions of the trigonal bipyramidal geometry. This position of lone pairs involves minimum repulsions as compared to axial position. The bond angle between two axial B-atoms is less than 180° and between equatorial and axial atoms is less than 90° . Example, ClF_3 , BrF_3 , IF_3 , ICl_3 etc. All these molecules have T-shaped geometry.

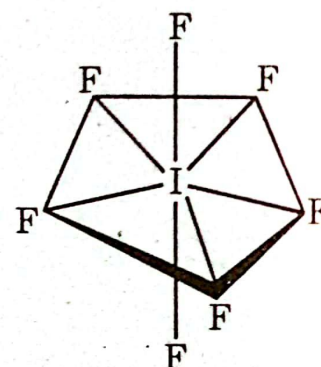


Q4

T-shaped geometry of ClF_3 molecule

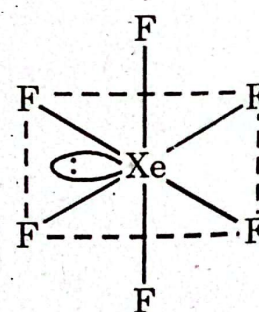
Q5

Q5



A diagram of a regular hexagonal prism. A vertical line passes through the center of the prism, with a force F acting downwards at the top and another force F acting upwards at the bottom. At each of the six vertices of the top hexagonal face, a force F is applied, directed outwards from the center of the face. Similarly, at each of the six vertices of the bottom hexagonal face, a force F is applied, directed inwards towards the center of the face.

A diagram of a regular hexagonal prism. A central point 'I' is located at the geometric center of the prism. Six forces, each labeled 'F', are applied at the vertices of the top hexagonal face. The forces are directed radially outward from the center 'I' to each of the six vertices. The forces at the top-left and top-right vertices are directed upwards and outwards. The forces at the bottom-left and bottom-right vertices are directed downwards and outwards. The forces at the top-left and bottom-left vertices are directed outwards to the left. The forces at the top-right and bottom-right vertices are directed outwards to the right.



The diagram shows the Lewis structure of XeF_6 . The central Xenon (Xe) atom is bonded to six Fluorine (F) atoms. Two bonds are axial (top and bottom), and four are equatorial (forming a square). All bonds are represented by solid lines. A lone pair of electrons is shown on the left side of the Xe atom, enclosed in an oval. Dashed lines extend from the four equatorial F atoms, and a dashed line extends from the bottom axial F atom, indicating the geometry of the molecule.

Significance of Spectrochemical series



Spectrochemical series can be used to predict the relative d-orbital splitting of two complexes containing the same metal ion with the different ligands. Similarly, the relative frequency of the absorption band can be predicted.

what is spectrochemical series?

Spectrochemical series is defined as the experimentally determined series on the basis of the strength of ligand field for the splitting of d-orbital of metal ion of complexes.

define spectrochemical series

The arrangement of ligands in a series in the order of increasing crystal field splitting is called spectrochemical series.

19 DISTRIBUTION OF ACTINIDES IN NATURE

Thorium and uranium are the most abundant actinides in nature with the respective mass concentrations of $1.6 \times 10^{-3}\%$ and $4 \times 10^{-4}\%$.

Uranium mostly occurs in the Earth's crust as a mixture of its oxides in the minerals uraninite, which is also called pitchblende because of its black color. There are several dozens of other uranium minerals such as *carnotite* ($\text{K}_2\text{UO}_2(\text{VO}_4)_2 \cdot 6\text{H}_2\text{O}$) and *autunite* ($\text{Ca}(\text{UO}_2)_2(\text{PO}_4)_2 \cdot n\text{H}_2\text{O}$). The isotopic composition of natural uranium is ^{238}U (relative abundance 99.2742%), ^{235}U (0.7204%) and ^{234}U (0.0054%); of these ^{238}U has the largest half-life of 4.51×10^9 years. The worldwide production of uranium in 2009 amounted to 50,572 tonnes, of which 27.3% was mined in Kazakhstan. Other important uranium mining countries are Canada (20.1%), Australia (15.7%), Namibia (9.1%), Russia (7.0%), and Niger (6.4%).

The most abundant thorium minerals are thorianite (ThO_2), thorite (ThSiO_4) and monazite, $(\text{Th, Ca, Ce})\text{PO}_4$. Most thorium minerals contain uranium and vice versa; and they all have significant fraction of lanthanides. Rich deposits of thorium minerals are located in the United States (440,000 tonnes), Australia and India (~300,000 tonnes each) and Canada (~100,000 tonnes).

4.7 CHELATING LIGANDS AND CHELATES

A chelating ligand is a bidentate or polydentate ligand which is attached to the same central metal atom by two or more of its donor atoms resulting in the formation of a complex having a strain-free ring structure. The complex having the ring structure is called chelate or chelated complex. The chelate is also called by various other names like *cyclic complex*, *ring-type complex* etc. The formation of a chelate is called *chelation* or *cyclisation*.

Factors affecting the stability of chelates

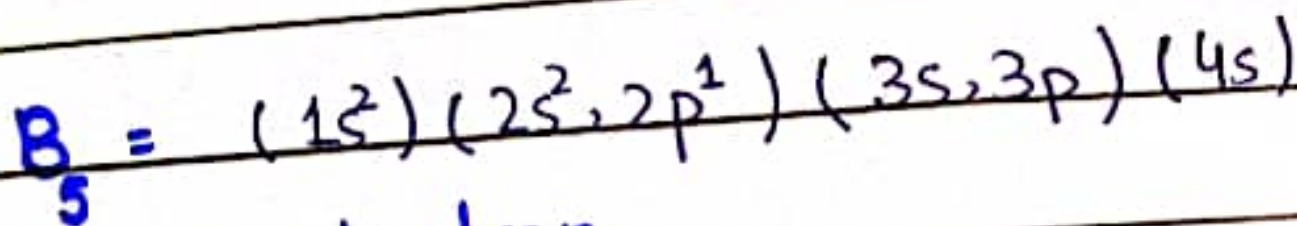
Following are important factors which influence the stability of chelates.

1. **Size of the chelate ring.** Chelates having 6-membered rings including the metal atom are more stable than those having 5-membered rings which in turn, are more stable than the chelates with 4-membered rings and so on.
2. **Number of chelate rings.** Greater in the number of chelate rings, greater is the stability of the chelate.
3. **Resonance effects.** Resonance enhances the stability of the chelate.
4. **Chelate effect.** The chelated complexes are known to be more stable than the non-chelated complexes. This effect is known as chelate effect.

Date: _____

Question (1x)

Effective Nuclear Charge of
valence p-electron in boron.



For p-electron

$$\text{Contribution by electron in same group} = 0.35 \times 2 = 0.7$$

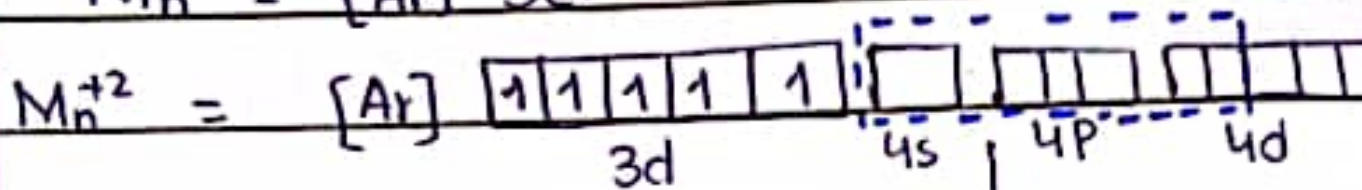
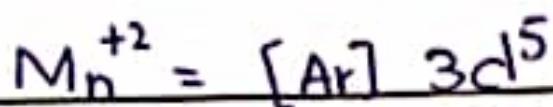
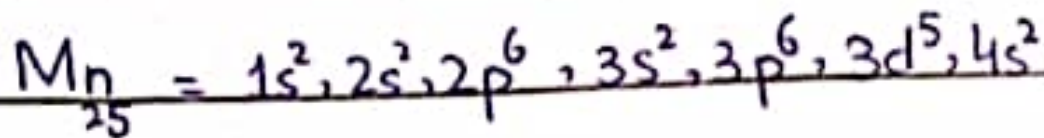
$$\text{Contribution by electron in } (n-1)\text{shell} = 0.85 \times 2 = 1.7$$

$$\text{Contribution by all inner } \bar{e} = 1 \times 0 = 0$$
$$0.7 + 1.7 = 2.40$$

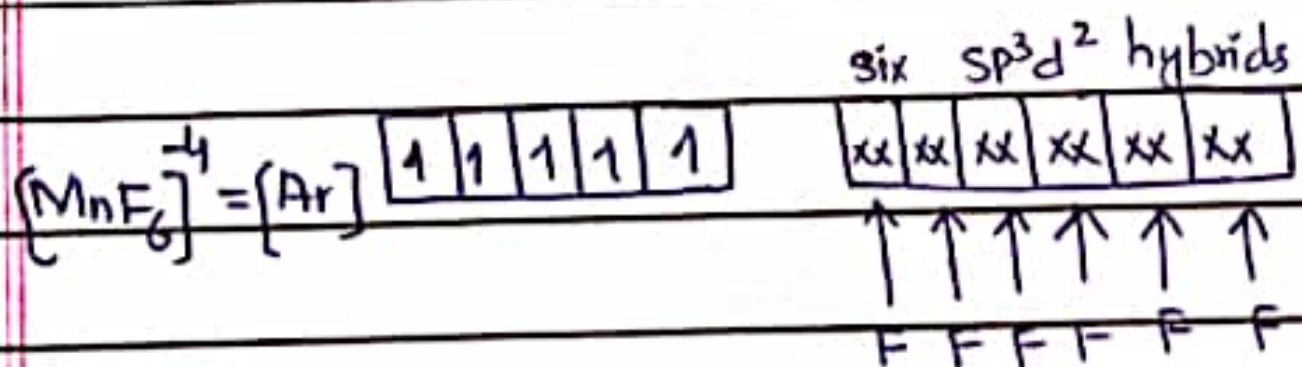
$$Z_{\text{eff}} = 5 - 2.40$$

$$Z_{\text{eff}} = 2.60$$

Structure of $[\text{MnF}_6]^{-4}$ on the basis of VBT.



hyberdize



$[\text{MnF}_6]^{-4}$ is paramagnetic due to presence of five unpaired e^- . The vacant $4d_{x^2-y^2}$ and $4d_{z^2}$ orbitals are hybridized with vacant 4s and 4p orbitals.